

DIVYANSHU KUMAR

AI Engineer | Full-Stack · RAG Pipelines · LLM Systems · Python

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PROFESSIONAL SUMMARY

ECE engineer and published researcher building at the hardware-software intersection — from Spring Boot microservices and LLM-powered RAG pipelines to YOLOv8 vision systems and autonomous robotics. Two production internships shipping real systems with measurable outcomes: 40% latency reduction via RAG, sub-200ms LLM roundtrip via prompt orchestration, and 90.5% mAP on a custom YOLOv8n model. Comfortable owning full AI systems end-to-end, from data pipeline to deployment. AWS certified, 500+ DSA problems solved.

KEY SKILLS

- **Generative AI & LLMs:** RAG pipelines, prompt engineering, LLM integration (Claude API / Anthropic SDK), agentic AI workflows, Anthropic MCP, Hugging Face (transformers), NLP
- **Vector Search & Embeddings:** FAISS vector database, sentence-transformers, semantic search, embedding pipelines
- **AI Frameworks:** Hugging Face Agentic AI, Anthropic MCP, TinyML (TFLite), PyTorch, OpenCV
- **Python & Backend:** FastAPI, Spring Boot, Node.js, Express.js, Python async, REST APIs, Microservices, WebSocket
- **Databases & Infrastructure:** PostgreSQL, MongoDB, Redis, Docker, Docker Compose, GitHub Actions CI/CD
- **Cloud & Security:** AWS Solutions Architect Associate, JWT/OAuth2, cloud architectures
- **ML / Vision:** YOLOv8 (custom training, 90.5% mAP), Web Serial API, ESP-NOW, Arduino, ESP32

PROFESSIONAL EXPERIENCE

Infosys Springboard

Nov 2025 – Jan 2026

Java Full Stack Developer Intern (AI Track)

Remote

- Designed and deployed an LLM-powered RAG pipeline (Python, sentence-transformers, FAISS, Claude API) translating legal queries into semantic search over a vector knowledge base; reduced resolution latency by **40%**.
- Implemented prompt engineering strategies for LLM query decomposition and context retrieval; applied retry and circuit-breaker patterns for production-grade AI pipeline reliability.
- Integrated external APIs (NGO Darpan, Bar Council) with JWT/OAuth2 role-based access; authored end-to-end AI pipeline documentation adopted by the full team.

CertifyO

May 2025 – Jul 2025

Full Stack Developer Intern

Remote

- Engineered **20+ RESTful API endpoints** (Node.js, MongoDB) with query optimisation, improving data throughput by **40%**; implemented Redis session caching reducing API latency by **60%** and backend load by **35%**.
- Scaled real-time WebSocket platform to **500+ concurrent sessions**; profiled and resolved event-loop bottlenecks cutting p99 latency by **35%** at 99.8% uptime.

PROJECTS

HexMind – Agentic AI Co-Pilot | Python, FastAPI, Claude API, Anthropic MCP, React.js

Mar – May 2026

- Built an agentic AI orchestration pipeline using Claude API and Anthropic MCP; engineered structured prompt templates translating natural language to JSON firmware commands with sub-**200ms** LLM roundtrip over USB serial.
- Integrated YOLOv8n vision pipeline (**90.5% mAP@0.5**) for real-time obstacle context injection into LLM prompts; implemented session-aware multi-turn agentic state management — research published at **NCETST-2026**.

Road Hazard Detection System | Python, FastAPI, YOLOv8, PostgreSQL, Redis, Docker

Feb 2026

- Deployed YOLOv8-powered containerized detection backend with Docker Compose; used Redis pub/sub for async alert pipeline handling **500+ concurrent clients at sub-100ms latency**.
- Implemented health-check endpoints, structured logging, and pytest/JUnit integration tests; configured CI/CD via GitHub Actions ensuring reliable AI solution deployment and monitoring.

EDUCATION

Bihar Engineering University – GEC West Champaran

2022 – 2026

B.Tech, Electronics and Communication Engineering

CGPA: 8.10

- Coursework: DSA, DBMS, Operating Systems, OOP, Computer Networks, Software Engineering
- Published researcher (NCETST-2026) | 500+ competitive programming problems (LeetCode/GFG)

CERTIFICATIONS

AWS Solutions Architect Associate | **Anthropic** Advanced MCP, Agent Skills & AI Fluency (2026) | **Oracle** AI Foundations

Hugging Face Agentic AI | **J.P. Morgan** Software Engineering Virtual Experience